

Task 14.3 Hospital Operations Dashboard – SQL-Based Analysis Report

1. Overview

This report presents an analytical study of hospital operations using structured query language (SQL). The objective is to evaluate key operational aspects such as **bed utilization, patient wait times, staff availability, and overall service efficiency**.

The analysis is performed on a hospital dataset containing operational, clinical, and administrative attributes. Instead of visualization tools, all insights are derived using SQL queries, and corresponding query results are included as snapshots for validation.

2. Objective of the Analysis

The primary goals of this analysis are:

- To measure **hospital efficiency and performance**
 - To evaluate **patient handling capacity and workload**
 - To assess **staff availability in relation to demand**
 - To analyze **waiting time impact on service quality**
 - To derive insights supporting operational decision-making
-

3. Dataset Description

The dataset includes the following categories of information:

- **Hospital Information:** ID, Name, Type, Sector, Location
- **Infrastructure:** Total Beds, ICU Beds, Operation Theatres
- **Staffing:** Doctors, Nurses, Paramedical Staff
- **Patient Metrics:** Outpatients, Admissions, Emergency Cases

- **Operational Metrics:** Waiting Time, Length of Stay, Surgeries
- **Quality Indicators:** Satisfaction Score, Mortality Rate, Infection Rate

This structured dataset enables comprehensive SQL-based analysis of hospital operations.

4. Key Performance Indicators (KPIs)

	Avg_Waiting_Time_Minutes	Total_Patient_Load	Total_Staff	Estimated_Bed_Utilization	Avg_Length_of_Stay_Days	Avg_Patient_Satisfaction
▶	92.94	14723539	4204392	0.97	5.71	7.3

The following KPIs were derived using SQL queries:

4.1 Average Waiting Time

- Measures the average time patients wait for services.
 - Indicates efficiency of hospital processes.
 - Lower values represent better operational performance.
-

4.2 Total Patient Load

- Calculated as the sum of:
 - Daily Outpatients
 - Monthly Admissions
 - Emergency Cases
 - Represents overall demand on hospital resources.
-

4.3 Staff Availability

- Total count of:
 - Doctors
 - Nurses
 - Paramedical Staff
- Helps evaluate workforce capacity.

4.4 Estimated Bed Utilization

- Derived using:
 - $\text{Monthly Admissions} \div \text{Total Beds}$
 - Acts as a proxy for bed occupancy due to absence of real-time occupancy data.
-

4.5 Average Length of Stay

- Indicates how long patients remain admitted.
 - Higher values may suggest inefficiencies or complex cases.
-

4.6 Patient Satisfaction Score

- Reflects overall patient experience and service quality.
-

5. SQL-Based Analytical Queries

All KPIs were computed using SQL aggregation functions such as:

- `AVG()` for averages
- `SUM()` for totals
- `ROUND()` for precision
- Conditional counting for categorical insights

Snapshots of query results are included to demonstrate outcomes and validate calculations.

6. Key Analytical Insights

Based on SQL query results, the following insights were observed:

	Hospital_Type	Total_Beds	Total_Admissions
▶	Specialized	1471165	1419014
	General	1396444	1364498
	Teaching	1411862	1369679

6.1 Patient Load vs Capacity

- High patient load compared to available beds indicates **potential overcrowding**.
 - Suggests need for better resource allocation.
-

6.2 Waiting Time Impact

	Hospital_Name	Waiting_Time_Minutes	Patient_Satisfaction_Score
▶	Hospital_VAXQQ	5	7.3
	Hospital_PDDVG	5	5.1
	Hospital_LKERU	5	5.6
	Hospital_CTSHP	5	5.2
	Hospital_NWZHD	5	5.5
	Hospital_WVYOX	5	8
	Hospital_EAONI	5	9

- Increased waiting time negatively impacts **patient satisfaction**.
 - Highlights inefficiencies in service delivery.
-

6.3 Staff Sufficiency

	Hospital_Name	Total_Staff	Patient_Load
▶	Hospital_SUXWV	1468	3124
	Hospital_OKGWR	1461	3219
	Hospital_CJHMO	1448	2000
	Hospital_EOLNC	1428	1801
	Hospital_TSDGA	1399	3834
	Hospital_GZSTK	1398	2920
	Hospital_VNBGV	1361	3676

- Comparison of staff count with patient load shows:
 - Some hospitals may be **understaffed**
 - Others operate efficiently with balanced ratios
-

6.4 Operational Efficiency

	Hospital_Type	Avg_Wait_Time
▶	General	93.97
	Teaching	93.09
	Specialized	91.78

	Sector	Avg_Stay
▶	Public	5.73
	Private	5.69

	Hospital_Type	Total_Surgeries
▶	Specialized	372778
	General	370348
	Teaching	375011

- Length of stay helps identify:
 - Efficient hospitals (shorter stays)
 - Resource-heavy hospitals (longer stays)
-

6.5 Emergency Preparedness

	Emergency_Department	Hospital_Count
▶	Yes	2721
	No	2779

- Availability of emergency departments indicates readiness to handle critical cases.

7. Limitations of Analysis

- Actual **bed occupancy data** was not available; estimated using admissions.
- No real-time or time-series data for trend analysis.
- Insights are based purely on SQL aggregation without visual representation.

8. Conclusion

The SQL-based analysis successfully evaluates hospital operations across key performance areas. The derived KPIs provide meaningful insights into:

- Resource utilization
- Patient handling efficiency
- Staff adequacy
- Service quality

This approach demonstrates how SQL alone can be effectively used for **data analysis and decision support**, even without visualization tools.

9. Summary

This project highlights the practical application of SQL in analyzing healthcare operations. The insights derived can assist in improving hospital efficiency, optimizing resources, and enhancing patient experience.

